Historical source

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Continues an entry from an earlier page.

be made to assume the appearance either of an acid salt or metal, at the pleasure of the operator; and therefore has been considered both as a falne substance and as a semi-metal. It is not known at what time this mineral was discovered; though, as it abounds in many different kinds of ores, it is probable that the pernicious properties it manifetts would very soon make it be taken notice of by metallurgifts. Ariftotle makes mention of a substance called ; and his difcipile Theophrastus makes mention of one named , which by Dioscorides and others was called about the beginning of the Christian era. By this, however, it appears, that they only meant the fubftances now called fandarach and orpiment; and Avicenna who lived in the 11th century, is the first who expressly mentions white arsenic, as well as its sublimate. It is not known by whom arsenic was first reduced to a metallic form. Paracelsus affirms that arsenic, sublimed with egg-shell lime becomes like silver; and, in 1675, M. Lemeri makes mention of a method of fubliming arsenic with fixed alkali and fepa.

The true nature of arsenic being thus so little known, it is no wonder to find chemists differing very much as to the class of natural bodies in which it ought to be placed. Avicenna and a great number of others class it with the sulphurs; Albertus Magnus and his followers, among the salts. Becher considers it as a kind of foam, or saline sulphureous body. Later experiments, however, have made it evident, that white arsenic consists of an acid united to phlogiston; and that by diminishing the latter, the acid becomes more and more apparent; while, on the contrary, by augmenting the quantity of phlogistic matter, the arsenic assumes the metallic form. With respect to the sulphureous nature of arsenic, it appears, indeed, that the regulus itself, as well as orpiment and realgar, are inflammable substances; but it is not so with white arsenic. This inflammability, therefore, which arsenic in a certain state has in common with zinc and several other substances, will not denominate it sulphureous, any more than those of other bodies which possess the common property of inflammability can be denominated sulphureous.

It is commonly said, that arsenic mineralizes metals ; and therefore, says Mr Bergman, it is considered as a sulphur by some, who yet extend the idea of mineralization so far, as under it to comprehend all mixtures of which metals make a part. But if we examine this signification a little more accurately, we shall fee that it is extended too far ; for if this be admitted, we must at the same time allow, that no native metal is to be found. Thus the gold called native, is seldom if ever found pure, but more or less mixed with copper or silver ; and so with other metals. If, therefore, arsenic, which, unless in its reguline state, never dissolves other metals, be considered as a mineralizing sulphur, what hinders us from saying that gold is mineralized by silver or copper, and in general every metal mineralized by some other ? It is much more natural to suppose that those metals are mineralized which are actually dissolved and concealed by a menstruum. Sulphur is the chief agent employed by nature for this purpose ; and though the acids of vitriol, phosphorus, nitre, and sometimes even the aerial acid, occasion the metals to put on an appearance foreign to their nature, yet the

number of these is so small, that compared with the sul- phurated minerals, they almost vanish.

This mineral, so troublesome to the mineralogist, occasioned the alchemists to suspect the existence of a certain arsenical principle indispensably necessary to the perfection of every metal. Even as late as 1773 a question to this purpose was proposed by the Royal Academy of Sciences at Berlin: the prize was adjudged to M. Monnet, who in his answer considered arsenic as a semi-metal of a peculiar kind, which is so far from constituting any essential part of metals, that its presence is always attended with inconveniences, either by carrying off the metal as it flies away, or spoiling the moulds in which it remains. These considerations, however, do not hinder us from asserting that the acid of arsenic, like others, is a mineralizing substance, if at any time it happens to meet with metals in the bowels of the earth, and to unite with it in that form.

Arsenic in its pure state is well known to be a most destructive and deadly poison, for which the art of medicine has scarcely as yet afforded a cure. Mr Bergman is of opinion that it acts as an highly corrosive acid, even when applied externally. He also tells us, that the dry acid is more destructive than white arsenic; the regulus and realgar left to. From an experiment of Mr Scheele, however, in which eight grains of arsenical acid were given to a cat, it does not appear that it acts more violently than white arsenic. The extreme danger attending this substance when taken into the human body, arises from its infusibility, and the difficulty of decomposing it; for there can but little danger arise from a liquid, unless like corrosive acids, it should at once burn the substance of the stomach like fire; or, like laurel water suspend the action of the nervous system. Corrosive sublimate, solutions of mercury

in aquafortis, &c. will as certainly poison as arsenic; but they are much less difficult to cure, because any alkaline substance will certainly decompose them and destroy their deleterious efficacy. Arsenic, on the contrary, cannot be decomposed, nor united with any known substance, at least in such a short time as the exigence of the case we speak of would require, without a considerable degree of heat. It therefore remains in the stomach, continually exerting its mischievous qualities, unless it can be discharged by vomiting.

The symptoms attending arsenic when swallowed are, nausea, fickenels, and retching to vomit, about half an hour after it is taken. These are followed by violent vomitings, hiccups, and pains in the stomach and bowels. Convulsions and palies of the limbs frequently succeed, with intense heats, cold sweats, palpitations of the heart, extreme anxiety, prostration of strength, thirst and dryness of the mouth and throat, loss of reason, and at last death. If the quantity taken was considerable, the patient dies in seven or eight hours after taking it; and the stomach and intestines are found, upon dissection, to be corroded and perforated. When this is not the case, violent putrefactive symptoms soon ensue after arsenic is swallowed; for the bodies of those who are poisoned by it generally have abundance of red or purple spots even before death. It remarkably inflames the coats of the stomach, and the putrefaction

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